A GROUP SCHEME OVER A FIELD HAS AN OPEN-AND-CLOSED SUBSCHEME THAT IS A COUNTABLE UNION OF AFFINES

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Allowed standard prerequisites (editorial).

- Lemma lemma-group-scheme-over-field-open-multiplication (canonical tag 047K). cited at proof lines 1430-1431. The n-slot multiplication maps m_n are open.
- Lemma lemma-open-subgroup-closed-over-field (canonical tag 047S). cited at proof lines 1432-1434. An open image of a homomorphism of group schemes over a field is closed.

Source reference. Original lemma 047K: sources/stacks-chow/tags.txt, source lines 5511–5511. Cited at proof line 1430-1431: openness of each m_n makes G' an open subgroup scheme; load-bearing.

Source reference. Original lemma 047S: sources/stacks-chow/tags.txt, source lines 5519–5519. Cited at proof lines 1432-1434: upgrades G' from open to closed subgroup scheme; load-bearing.

Lemma 0.2. *Let G be a group scheme over a field. There exists an open and closed subscheme $G' \subset G$ which is a countable union of affines.*

Proof. Let $e \in U(k)$ be a quasi-compact open neighbourhood of the identity element. By replacing U by $U \cap i(U)$ we may assume that U is invariant under the inverse map. As G is separated this is still a quasi-compact set. Set

$$G' = \bigcup\nolimits_{n \geq 1} m_n(U \times_k \ldots \times_k U)$$

where $m_n : G \times_k \ldots \times_k G \to G$ is the n-slot multiplication map $(g_1, \ldots, g_n) \mapsto m(m(\ldots(m(g_1, g_2), g_3), \ldots), g_n)$. Each of these maps are open (see Lemma 047K) hence G' is an open subgroup scheme. By Lemma 047S it is also a closed subgroup scheme. □

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